A Comparative Analysis of File System Type and Digital Forensic Data Recoverability

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***Abstract*—Digital forensic investigations often require recovering deleted or reformatted data from storage media. Existing research has examined file-carving techniques, recovery tools, and individual file systems. However, comparatively less research has directly evaluated how different file-system architectures affect data recoverability under identical experimental conditions. This study proposes a controlled, reproducible comparison of forensic file recovery across NTFS, exFAT, ext4, Btrfs, and bcachefs using a standardized NixOS virtual-machine environment.**

This study will experimentally evaluate whether file-system architecture affects forensic data recoverability under controlled conditions. A reproducible NixOS virtual-machine environment will be used to test NTFS, exFAT, ext4, Btrfs, and bcachefs using fresh virtual disks for each experimental run. A standardized dataset containing known files will be written to each file system and cryptographic hashes will be recorded before a controlled data-loss procedure is performed and file recovery will then be attempted using PhotoRec.

This study aims to determine whether file-system type produces measurable differences in forensic data recoverability when hardware-related factors are intentionally controlled. The resulting virtual test environment is intended to provide a reproducible baseline for comparing file-system behaviour.

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

Write your introduction here...

II. Literature Review

Write your previous work here. We explore related work in three broad areas: theme 1, theme 2, and theme 3, followed by a literature review matrix that highlights the unanswered research gaps in each theme.

A. Literature Review Matrix

B. Theme 1

C. Theme 2

D. Theme 3

III. Methodology

Explain the comprehensive approach you used to conduct this research here.

A. Research Method

B. Data Sampling

C. Data Analysis Techniques

D. Data Measurement Models

IV. Evaluation

A. Research Questions/Hypothesis

B. Evaluation metrics

- Metric 1
- Metric 2
- Metric 3
- .....

C. Result

Provide the result of your analysis in the form of figures and tables here and explain how your results answer your research questions

V. Discussion

What challenge did you face? Is your result generalizable? How did you validate your result?

VI. Conclusion and Future Work

What are the existing gaps and unanswered research questions? what is the next step?

Acknowledgment

References

Provide your citations as long as you continue with your research. For example, Einsteign et.al. [1] demonstrated that energy converts...

References

[1] A. Einstein, "On the electrodynamics of moving bodies," *Annalen der Physik*, vol. 17, pp. 891–921, 1905.